Jonathan Frankle and Michael Carbin. 2018. The Lottery Ticket Hypothesis: Training Pruned Neural Networks. *arXiv:1803.03635 [cs]*. ArXiv: 1803.03635.

Mario Giulianelli, Jack Harding, Florian Mohnert, Dieuwke Hupkes, and Willem Zuidema. 2018. Under the Hood: Using Diagnostic Classifiers to Investigate and Improve how Language Models Track Agreement Information. *arXiv:1808.08079 [cs]*. ArXiv: 1808.08079.

Yova Kementchedjhieva and Adam Lopez. 2018. *Indicatements* that character language models learn English morpho-syntactic units and regularities. In *Proc. of Workshop on Analyzing and interpreting neural networks for NLP*.

Urvashi Khandelwal, He He, Peng Qi, and Dan Jurafsky. 2018. Sharp Nearby, Fuzzy Far Away: How Neural Language Models Use Context. *arXiv:1805.04623 [cs]*. ArXiv: 1805.04623.

Tal Linzen, Emmanuel Dupoux, and Yoav Goldberg. 2016. Assessing the Ability of LSTMs to Learn Syntax-Sensitive Dependencies. *arXiv:1611.01368 [cs]*. ArXiv: 1611.01368.

Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. 2013. Distributed Representations of Words and Phrases and their Compositionality. In *Advances in Neural Information Processing Systems 26*, pages 3111–3119. Curran Associates, Inc.

Ari S. Morcos, Maithra Raghu, and Samy Bengio. 2018. Insights on representational similarity in neural networks with canonical correlation. *arXiv:1806.05759 [cs, stat]*. ArXiv: 1806.05759.

Matthew E Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. 2018. Deep contextualized word representations. *arXiv preprint arXiv:1802.05365*.

Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. 2017. SVCCA: Singular Vector Canonical Correlation Analysis for Deep Learning Dynamics and Interpretability. *arXiv:1706.05806 [cs, stat]*. ArXiv: 1706.05806.

Ravid Shwartz-Ziv and Naftali Tishby. 2017. Opening the Black Box of Deep Neural Networks via Information. *arXiv:1703.00810 [cs]*. ArXiv: 1703.00810.

Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. 2016. Understanding deep learning requires rethinking generalization. *arXiv preprint arXiv:1611.03530*.

Kelly W. Zhang and Samuel R. Bowman. 2018. Language Modeling Teaches You More Syntax than Translation Does: Lessons Learned Through Auxiliary Task Analysis. *arXiv:1809.10040 [cs]*. ArXiv: 1809.10040.

A Performance Out Of Domain

Because SEM tags and PTB POS tags were both trained on the GMB corpus, we present the SVCCA similarities on an in-domain GMB test corpus as well as the Wikipedia test corpus used elsewhere in the paper. The results are in Figures 10-11. In general correlations are higher using the original tagging domain, but not enough to contradict our earlier analysis. The shapes of the curves remain similar.

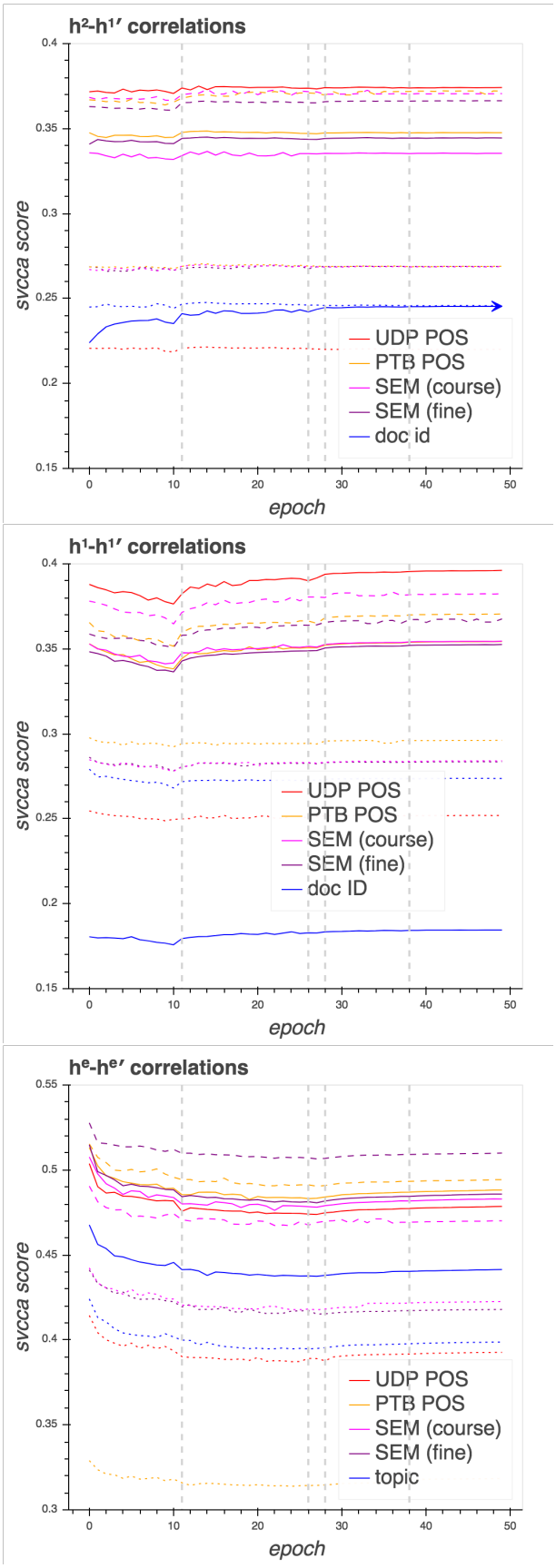

Figure 10: SVCCA correlation scores between LM and y_{t+1} tag predictor. Dotted lines use models trained on randomly shuffled the data. Dashed lines use GMB domain test data.

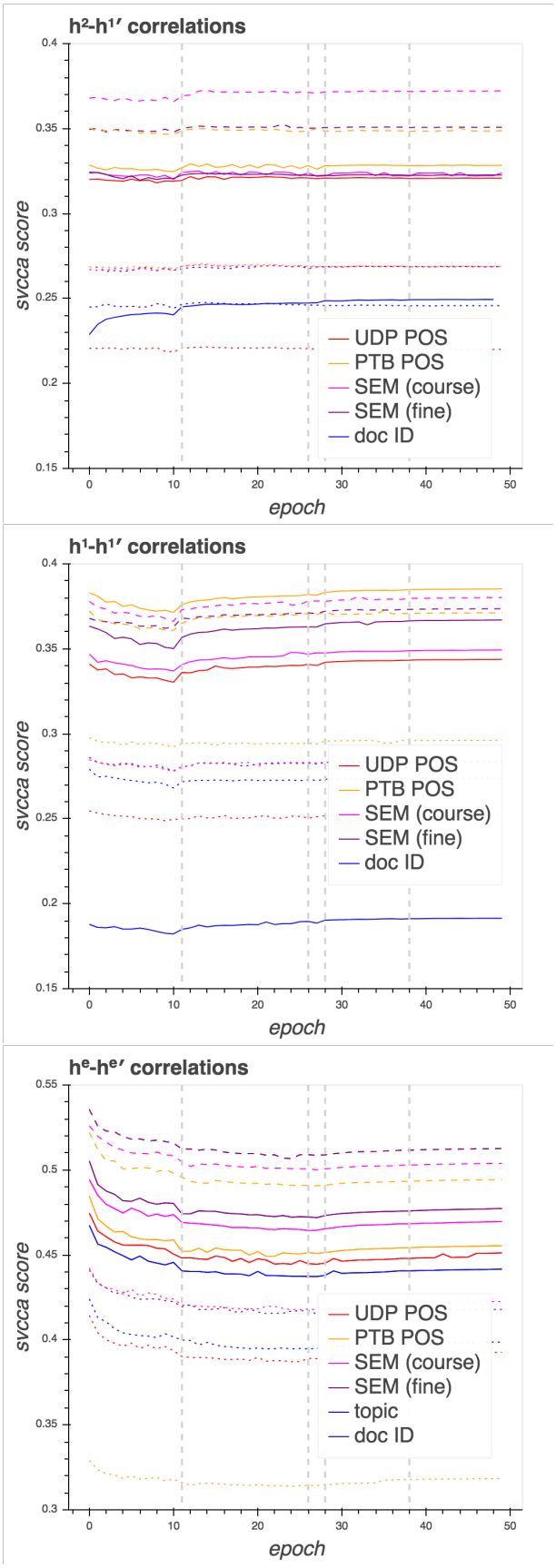

Figure 11: SVCCA correlation scores between LM and y_t tagger. Dotted lines use models trained on randomly shuffled the data. Dashed lines use GMB domain test data.